

PHILOSOPHY OF MAN

INTRODUCTION TO PHILOSOPHY OF MAN

“Every human being must philosophize because first of all philosophy is everybody’s business; every time we reason, we use philosophy.”- **Aristotle**

Why do we need to study Philosophy?

- It helps us clarify issues, discriminate among options and make better decisions. Philosophy helps us to choose the better choice or options.
- Philosophy has a practical side. Philosophy is something that we can do. It helps us to be critical and with the help of philosophy, we protect ourselves from destructive ideas.
- Philosophy can be a pleasurable and inspirational element in our life. It helps us to recognize the truth and the false, real and unreal.
- Philosophy assists us to the truth of commitment and that commitment involves faith. As according to Samuel Butler “You can do very little with faith but you can do nothing without it.” This means that thinking should guide our faith not faith guiding our thinking.

Techniques in studying Philosophy

- remain open-minded
- be perceptive
- be critical

Philosophical Attitude

- Tolerance- respecting the idea of others/ openness.
- In philosophy, we use dialogue which philosophers called great conversations. Philosophy is not something one simply learns, it is also something one lives. It is not only learned in the classroom but more on reflections.

Ways of learning Philosophy

- Consider the thoughts of other ordinary people.
- Consider the reflections of other ordinary people about life.

The risks of Philosophy

- Risk of failure
- Risk of direct self-awareness
- Risk of disturbance of early pattern or action or thought. (Doors once open are hard to close- when you learn or know something, it is hard to be unlearned.)

The adventure of Philosophy

- According to Albert North Whitehead “Philosophy is an adventure of ideas” and that little knowledge is a dangerous thing. When we say adventure, we start with little knowledge, then more knowledge to greater knowledge.

Definition/Meaning of Philosophy

- Philosophy comes from the Greek words **philos (philos)** which means love and **sophia (sophia)** which means wisdom.
- It is a branch of human science which deals with the formal reasons and formal principles of things knowable through the natural light of human reason alone.
- It is a human science because philosophy is not just a gathering of knowledge but an organized body of knowledge.
- It deals with the proper reasons and formal principles of things because philosophy searches for what is essential and it is more on formal features rather than with the physical or material features of a thing.
- This makes philosophy different from the positive sciences such as Biology, Chemistry, and Physics etc. Positive sciences deal with the physical or material constituents and physical principles of things, while philosophy goes beyond the physical or the material features of a thing and it studies the nature of life and its principle.
- Take for example the definition of man.
 - **Positive science:** *“Man is a member of the Genus Homo, Family Hominidae, Order Primates, Class mammalia, characterized by erect posture and opposable thumb, distinguished by the ability to communicate by means of organized speech and to record information in a variety of symbolic systems”*
 - **Philosophy:** *“Man is a rational animal, that is, a being having an organic sensitive body and a rational soul, which through the specific rational faculties of reason and will is capable of abstract knowledge through discovery, inference or education, and is also capable of discretionary free action, of culture and progress, and of communicating with his kind by means of articulate voice and other meaningful signs.”*
- Knowable through the natural light of human reason alone because man alone has the capability to think and rationalize.
- According to **Blaise Pascal**, **“The heart has reason in which reason itself cannot understand”**
- **Philosophy** is the science of “things”.
- **Philosophy** includes everything as its object as seen in their final causes. It tries to go down into the foundational principles.
- **Philosophy** is also an attempt to reach a systematic knowledge of the different aspects of reality. It is an attempt to reach answers, and conclusions which can still be open for revisions.
 - Because philosophy is systematic, it should be orderly, consistent, comprehensive and logical knowledge. That is why we cannot understand something if it is disorderly.

Why does man philosophize?

- To understand ourselves better.
- Man is searching for wisdom.
- It is the cry of the human heart and mind.
- Man seeks answers to his questions.
- Man is intelligent.

Beginning of Philosophizing

- Wonder
- Doubt
- Limits situation (crisis)
- Metaphysical uneasiness- related with spiritual things-inner restlessness
- For Socrates, philosophy is knowing ourselves that is why Socrates would say “**Know Thyself**”. Philosophy is really a discovering of our own selves and knowing more of our own selves the basic questions in philosophy. Some of these basic questions are as follows:
 - **Who am I? What is Man?**
 - **Where am I going?**
 - **What is my future and destiny?**
 - **What should I do to make my life happy?**
- With this, we can conclude that philosophy is all about the **MEANING OF LIFE**. To see the meaning of life, the following questions must be asked again of ourselves:
 - **What is the supreme purpose and goal of human living?**
 - **What consists of man's perfection and happiness?**
 - **What is the ultimate objective of human strivings? Is it ever attainable?**
 - **What is the *summum bonum* of human existence?**
 - **What is the remedy and cause of human suffering?**

Branches of Philosophy

- **Logic**
 - Gk. logike (logike) which means word or thought.
 - The science and art of correct thinking.
- **Cosmology**
 - Gk. Kosmos (kosmos) which means universe.
 - A study on the universe or the physical world.
- **Psychology**
 - Gk. Psyche (psyche) which means mind or soul
 - On living beings, vital operations and the principle of life (soul).
- **Ethics**
 - Gk. ethos (ethos) which means morality.
 - On the morality of human acts.
- **Ontology/General Metaphysics**
 - Lt. ens or on which means being.
 - On beings in general.

- **Theodicy**
 - Gk. Theos (theos) which means God.
 - On the First Cause of universal contingent beings.
- **Epistemology**
 - Lt. episteme which means knowledge.
 - On the validity of human knowledge.
- **Anthropology**
 - Gk. Anthropon (anthropon) which means man.
 - A study about man.
- **Political Philosophy**
 - A study about the state or government.
- **Social Philosophy**
 - A study of man in relation with the society he lives in.
- **Aesthetics**
 - A study of art and beauty.
- **Axiology**
 - A study of human values

LOGIC

Meaning/definition of Logic

- Science and art of correct thinking.
- Is an **organon** meaning instrument. It is an instrument for discovering and presenting truths.
- It is through logical methods that sciences discover truths.
- The ability to reason is inherent to man. We have this so called **COMMON SENSE**. This is what we call Natural Logic and this guided man to the use of fire, of water etc.
- But common sense is prone to mistakes; we need therefore a systematized instruction on how to build up our natural talents and how to use them effectively.
- Therefore, we need the science and art of logic to develop the habit of confident reasoning.
- Logic is a **SCIENCE**.
- **SCIENCE** is a systematic body of knowledge.
- Logic is a science because it is a body of knowledge that is systematically arranged and confirmed to be true.
- Logic is an **ART**.
- **ART** is the habit of doing something with a certain degree of order and harmony. It is also a normative science which means it uses rules to be followed
- Logic is an **ART** because it is a practical science which leads to the formation of the habit of thinking easily, orderly and properly.
- Logic is the science and art of correct **THINKING**.
- **Thinking** refers to any or to all the movements of the mind such as imagining, recalling, memorizing, comprehending, analyzing, day-dreaming or solving problems.
- In logic, thinking means **inference**.

- **INFERENCE**- getting a truth or conclusion from something formerly known to the thinker.

2 Kinds of Inference

- **1. Inductive Inference**- draws a universal or general conclusion from a series of individual interrelated facts.
- **2. Deductive Inference**- draws a particular conclusion on the basis of assumed common or universal principles.
 - **Premises** (statements) - evidence.
 - **Conclusion** (statement) - what is claimed to follow from the evidence.
 - Examples:

All cats are animals.	-premise
Felix is a cat.	-premise
Therefore, Felix is an animal.	-conclusion

All salesmen are extroverts.	-premise
Mario is a salesman.	-premise
Therefore, Mario is an extrovert.	-conclusion

Why should we study logic?

- We begin our solution to this question by observing that everyone obviously desires to know. This self-evident assertion simply means that a human being is so constituted that he cannot help wanting to know. A human being is a "knowing being".
- All human beings then in varying degrees want to know why things are so. An obvious sign of this is that even as kids we often ask for the why of things. We are insatiably curious.
- Logic then is a tool in helping us find out why things are as they are. An axe is a tool for cutting down a tree. A sharp axe is an efficient instrument for cutting down a tree. The power of thinking is an instrument for knowing the why and wherefore of things but thinking sharpened by skill in logic is an efficient instrument for scientific knowing.

Ideogenesis (how is an idea being formed in the mind?)

- **Perception/Sensation→Apprehension→Abstraction = IDEA**
- To have an idea, there should be an encounter between the subject and the object. This is called **PERCEPTION or SENSATION**.
- **SIMPLE APPREHENSION**- an act of the mind by which we know the real meaning of a thing without affirming or denying anything about it. In this mental act, phantasms are being formed in our imaginations.
- **ABSTRACTION**- is the mental process for the formation of an idea which includes picking up from, or drawing out from.

3 Acts of the Mind

1. Simple Apprehension- act of the mind by which we understand the fundamental nature of a thing.

2. Judgment- act of the mind by which we compare two concepts and declare them to be either in conformity or disagreement with each other.

- **Proposition-** is a kind of sentence wherein the subject and the predicate are combined so as to emphasize something true or false.

- Examples.

1. Socrates is an ancient Greek philosopher.
2. Man is not an angel.
3. Persons are not things.

3. Reasoning- act of the mind by which we gain new truths from what is formerly assumed to be true.

- **Argument/Syllogism-** combining sentences or propositions so as to form a new judgment.

- Examples:

1. All philosophers are good thinkers.
Friedrich Wilhelm Nietzsche is a philosopher.
Therefore, Friedrich Wilhelm Nietzsche is a good thinker.
2. All men are rational beings.
Peter is a man.
Therefore, Peter is a rational being.

2 Logical Properties of an Idea

1. Comprehension- sum-total of the thought elements or conceptual features contained in an idea.

In giving the comprehension of an idea, we are simply giving the definition of the concept or idea.

- **Definition-** knowing the essence of a thing. Giving the specifying difference and the genus.

- Examples:

1. Man- rational animal.
2. Man- being that is substance, bodily, living, sentient and rational.

2. Extension- sum-total of individuals and categories of beings to which the idea may be applied.

- **Application:**

- When someone asks for the meaning of a term, he anticipates an answer from its comprehension.
- When a student asks for an example of a proposition, he anticipates an answer from its extension.

Example	
COMPREHENSION	EXTENSION
Substance	Spirits, minerals, plants, brutes, men
Material, substance	Minerals, plants, brutes, men
Living, material, substance	Plants, brutes, men
Sentient, living, material, substance	Brutes, men
Rational, sentient, living, material, substance	Men

What is the relationship between the comprehension and extension of an idea?

- They **VARY INVERSELY**. That is the greater the comprehension, the lesser the extension and the lesser the comprehension, the greater the extension.

TERMS

Meaning/Definition

Terms are verbal or written words or signs communicative of a concept or idea. The spoken word is an articulated human voice and the written word is the sign of the spoken word.

Sign is something that manifests a different thing aside from itself.

- Example:
 - Color red is a sign of martyrdom.

Kinds of Signs

1. Natural signs- signs whose connection with the objects they manifest is provided by nature.

- Examples:
 - A heavy dark cloud is** a natural sign of an oncoming rain.
 - Smoke** is a natural sign of fire.

2. Conventional signs- signs whose connection with the thing it manifests is provided by common understanding or agreement.

- Examples:
 - Flag
 - Traffic lights
- Spoken and written words**, therefore, are conventional signs. Thus, different people have different words to designate the same object.

Classification of Terms

A. According to Meaning

1. Univocal- term expressing the same meaning as applied to several subjects.

- Examples. Man, podium, stone

2. Analogous- term expressing cognate or related meaning.

- partially different, partially the same.

- Examples. Beautiful song- beautiful lady & Good mother- good heart.

3. Equivocal- term that is outwardly and externally identical or the same but expressing different meanings.

- Examples. son-sun, hot-hat-hut, bottle- battle

B. According to Comprehension

1. Concrete- these are terms that can be perceived; they are sensible and tangible.

- Examples: House, Church, Animal, Horse, River

2. Abstract- signifies a nature or quality though it exists on its own right and apart from the individual or subject.

- Examples :

- | | | |
|--------------|-----------|---------|
| ▪ Friendship | Humanity | Love |
| ▪ Manhood | Authority | Loyalty |

C. According to Extension

1. Singular- signifies one, definite, specific individual.

- **Signs and determinants:**

- Proper Nouns
- Nouns modified by adjective in the superlative degree
- Collective nouns
- Demonstratives
- Personal pronouns

- Examples: My father, His dog, The First gentleman, President George Bush, The highest mountain

2. Universal- signifies all the individuals within the extension of such concept.

- **Signs and determinants:**

- Universal expressions
- Universal ideas

- A statement which is true all the time.

- Examples: Everybody, All students , Nobody, Every, Each one

3. Particular- signifies a part or portion of the total extension of such concept.

- Term that stands for an indefinite part of an absolute extension.

- **Signs and determinants:**

- Indefinite pronouns or adjectives
- Use of numbers
- General propositions

- A statement which is true most of the time but not all the time.

- Examples: Some teachers, Few candidates, Most schools

D. According to Relation

1. Identical Terms- are those having the same comprehension and extension.

- Examples: lawyer and attorney, God and Supreme being, man and rational being, disciple and follower

2. Similar Terms- are those having the same extension but different comprehension.

- Examples: writer and journalist, teacher and professor

3. Compatible Terms- are those expressing features which may be present at the same time in one individual or subject.

- Examples: rich and humble, tall and dark and handsome , intelligent and beautiful, beautiful and intelligent

4. Incompatible Terms- are those expressing features which cannot be present together and simultaneously in one individual or subject.

- Examples: rich and poor, hot and cold, weak and healthy, good and bad

5. Relative Terms- are those that express a feature of a thing which cannot be thought of without implying another.

- Examples: master-slave, teacher-student, doctor-patient

6. Privative Terms- are those which express the absence or lack of perfection in an individual or subject.

- Examples: blindness- is the absence of sight, death- is the absence of life, ignorance- is the absence of knowledge

7. Contradictory Terms- are those so related that one is the simple negation of the other. There is no underlying middle between contradictory terms.

- Examples: mortal-immortal, something-nothing

8. Contrary Terms- are those that express the extreme opposites in a given category or series of the same class. There is an underlying middle between contrary terms.

- Examples: first and last, left and right

DEFINITION

Definition (2 elements)

Definiendum- term to be defined.

Definiens- defining term.

- Example:
 - Man is a rational animal.
 - man- *definiendum*
 - rational being- *definiens*

Types of Definition:

2 main divisions:

- **A. Nominal**
- **B. Real**
 - **Nominal-** expresses what the name means, not what a thing is.

-required when a new term is introduced into a vocabulary.

3 Kinds of nominal definition:

- 1. Ostensive/demonstrative definition-** indicates the meaning of a term by showing or pointing at the object.

-from the Latin word “ostendere” means to point.
-non-linguistic method of defining term.

- Examples.
 - Table
 - Ballpen
 - Waltz
 - Asterisk
 - Dollar

2. Synonymous- gives the same connotation of the tem.

- Examples:
 - Saccharides- Carbohydrates
 - Soleil- Sun

3. Etymological- gives the origin of the word.

- Example:
 - Acropobia (“acro”-high, “phobia”-fear) means the fear of height.
- **Real Definition-** tells us what the thing is, not the name means.
 - it digs deeper into nature of a thing.

2 subdivisions of real definition

1. Essential

2. Non-Essential

1. Essential- a definition that is constructed by genus & specific difference.

- Example:
 - Decagon polygon with ten sides.
 - The term “polygon” refers to a general class that contains decagons as a subclass; the term with “ten sides” refers to the distinguishing mark that sets it apart from other polygons.

2. Non- Essential (descriptive) definition- gives the more notable characteristics of a thing.

4 subdivisions of non-essential definition

1. Distinctive- (definition by property) gives the natural characteristic of a thing that follows the necessarily from the essence of the thing.

- Example:
 - Oxygen- is a gas 1.105 times as heavy as air.

2. Genetic- furnishes the mode of the origin of a thing.

- it describes how something is produced.

- Example:
 - Tsunami- is a long high sea wave cause by underwater earthquake.

3. Causal- describes a thing by efficient & final cause.

2 divisions of causal definitions

3.1. Definition by efficient cause- gives the producer of a thing.

- Example:
 - Web page- is a virtual page made by a web master.

3.2. Definition by final cause- gives the purpose or end on account of which a thing is produced.

- Example:
 - Sedative- is a drug for inducing sleep.

4. Accidental- explains a thing by giving characteristics that are contingently connected with it.

- Example:
 - The typhoon is tremendously strong.

Rules of a Good Definition

1. A definition should avoid vagueness & ambiguity.

Example:

Obscure & Figurative definitions

Example of obscure definition:

NET- is the reticulated fabric decussated at regular intervals with interstices & intersections.

Example of figurative definition:

LOVE- is a sweet misery.

A definition should not be circular.

Example:

A wall clock is a clock on the wall.

2. Definition should not be needlessly negative.

Example:

Odd is not even.

3. Definition must be precisised...it must not be too broad.

Example of definition that is too broad.

Bachelor- unmarried male.

4. Definition should not be too narrow.

Example of definition that are too narrow.

A woman is a married mother.

A doctor is a surgeon.

PROPOSITION

Judgment

- A mental operation that pronounces the agreement/disagreement between two ideas.
 - Example:

- The mind compares, a square & a polygon....recognizing the identity between these two ideas, it pronounces this judgment; "A square is a polygon.
- Or, comparing a table & a chair....., the mind enunciates that "a table is not a chair"
- As seen here, ideas are naturally ordained to judgment....simple apprehension is a preparation for the judgment of the intellect.

Truth of a Judgment

- Aristotle has this to say about truth....to say what is is not, or what is not is, is false; and to say that what is is, or that what is not is not, is true.....Truth, thus, is the conformity of the judgment to reality; falsity is its disagreementis important to note that it is not the mind that dictates what really should be; rather, it is reality that dictates what the mind should think....

Proposition

- A proposition is an expression of judgment. It may be something that is stated for the purpose of discussion or something to be dealt with as a statement of fact or truth.
- It is a statement in which something is affirmed or denied.
- A proposition is different from a sentence because a sentence is a word or a group of words expressing a complete thought.
- An example of a sentence is "RUN!" This is a single word expressing a complete thought and therefore a sentence.
- Therefore, all propositions are sentences but not all sentences are propositions.
 - "RUN!" is not a proposition because it does not express a judgment.

Categorical Proposition

- Elements of a Categorical Proposition
 1. Quantifier- part of the CP that tells the quantity of the proposition.
 - 2.) Subject and Predicate Terms- matter of the proposition.
 - Materials from which the proposition is made.
 - 3.) Copula- bonding verb- form of the CP.
 - Unifying principle that maintains the structure of the proposition.

Quality of a Proposition

1. Affirmative- the predicate is affirmed of the subject.
 - from the Latin word *affirmo* which means "I agree".
 - the first 2 vowels of the word are A and I. They are the affirmative propositions.
 - Example: PNOY is the president of the Philippines
2. Negative- the predicate is denied of the subject.
 - from the Latin word *negō* which means "I deny". The two vowels of the word are E and O. They are the negative propositions.
 - Example: GMA is not the President of the Philippines.

Quantity of a Proposition

1. Universal- the predicate is affirmed or denied of the whole of the subject.
 - Example: All mothers are loving parents.
2. Particular- the predicate is affirmed or denied of only part of the subject.
 - Example: Some politicians are corrupt officials.

4 Types of Categorical Propositions

1. Universal Affirmative (A)
 - All lawyers are politicians.
2. Universal Negative (E)
 - No lawyers are politicians.
3. Particular Affirmative (I)
 - Some lawyers are politicians.
4. Particular Negative (O)
 - Some lawyers are not politicians.

Logical Oppositions

▪ Contradiction/Contradictory Propositions

-when two propositions using the same subject and predicate terms oppose each other owing to their differences in both quantity and quality.

- A and O/ E and I

- Examples:

-“All men are mortal beings” and “Some men are not mortal beings” are contradictory propositions.

-“All men are not emotional” and “Some men are emotional” are contradictory propositions.

▪ Contrariety/Contrary Propositions

- when two universal propositions using the same subject and predicate terms but differ in quality.

- A and E

- Examples

-“All politicians are honest” and “No politician is honest” are contrary propositions.

-“No student is intelligent” and “All students are intelligent” are contrary propositions.

▪ Sub-Alternation/Sub-Alternating Propositions

- when two propositions using the same subject and predicate terms but differ in quantity with the same quality.

- A and I/ E and O

- Examples:

- “All men are liars” and “Some men are liars” are sub-alternating propositions.

- “Some students are not absent” and “No student is absent” are sub-alternating propositions.

▪ Sub-Contrariety/Sub- Contrary Propositions

- when two particular propositions using the same subject and predicate terms but differ in quality.
- I and O
- Examples:
 - “Some buildings are houses” and “Some buildings are not houses” are contrary propositions.
 - “Some criminals are not harmful” and “Some criminals are harmful” are contrary propositions.

Four Laws Governing Logical Oppositions

- Law of Contradiction
- Law of Contrariety
- Law of Sub-Alternation
- Law of Sub-Contrariety

Law of Contradiction

- Two contradictory propositions cannot be both true and both false at the same time.
 - * If A is true, O is false
 - * If O is true, A is false
 - * If E is true, I is false
 - * If I is true, E is false
 - * If A is false, O is true
 - * If O is false, A is true
 - * If E is false, I is true
 - * If I is false, E is true

Law of Contrariety

- States that two contrary propositions cannot be both true but they may be both false at the same time.
 - * If A is true, E is false
 - * If E is true, A is false
 - * If A is false, E is doubtful
 - * If E is false, A is doubtful

Law of Sub-Alternation

- a) States that the truth of the universal carries or implies the truth of the particular but not vice versa.
 - * If A is true, I is true
 - * If E is true, O is true
 - * If I is true, A is doubtful
 - * If O is true, E is doubtful
- b) States that the falsity of the particular carries or implies the falsity of the universal but not vice versa.
 - * If I is false, A is false
 - * If O is false, E is false
 - * If A is false, I is doubtful
 - * If E is false, O is doubtful

Law of Sub-Contrariety

- States that two contrary propositions cannot be both false but they maybe both true at the same time.

- * If I is false, O is true
- * If O is false, I is true
- * If I is true, O is doubtful
- * If O is true, I is doubtful

Rules in Translating Ordinary Statements into the Standard Form of Categorical Proposition

1. *Translate universal statements as universal propositions unless the statement points to a particular usage.*

- Examples:
 - Dogs bark.
→ All dogs are barking animals.
 - Filipinos are hospitable people.
→ Some Filipinos are hospitable people
 - Men are stronger than women.
→ Some men are people stronger than women.

2. *Add the missing complement to an adjective or to a describing phrase to show that they refer to classes/terms.*

- Examples:
 - All lions are fierce.
→ All lions are fierce animals.
 - Mothers love their children.
→ Some mothers are children lovers.

3. *Singular statements should be treated as universal statements.*

- Examples:
 - The first lady is very extravagant person.
→ The first lady is very extravagant person.
 - The author of Don Quijote is a good writer.
→ The author of Don Quijote is a good writer.
 - This student is not lazy.
→ This student is not a lazy guy.

4. *Quantifiers that refer to universal or particular should be replaced by: all, no or some respectively.*

- Examples:
 - Almost 50% of the students in logic are members of the debate team.
→ Some students in logic are members of the debate team.
 - Most of the teachers are nationalists.
→ Some teachers are nationalists.
 - Every student is a learner.
→ All students are learners.

5. *Exclusive statements should be translated into universal statements by reversing the order of the original statement.*

- Examples:
 - None but men are priests.
→ All priests are men.

- Only voters are citizens.
→ All citizens are voters
- 6. **Exceptive statements should be translated to an E statement.**
 - Examples:
 - All except seminarians are members of the club.
→ No seminarian is a member of the club.
 - Everybody except the judges are members of the administration
→ No judge is a member of the administration.
- 7. **“Not all” should be translated as an O proposition.**
 - Examples:
 - Not all applicants are qualified workers.
→ Some applicants are not qualified workers.
 - Not all politicians are liars.
→ Some politicians are not liars.

EDUCTION

Definition/Meaning of Eduction

Eduction- the formulation of a new proposition by the interchange of the subject and the predicate of an original proposition and/or by the use or removal of negatives.

Kinds of Eduction

1. **Conversion**
 2. **Obversion**
 3. **Contraposition**
 4. **Inversion**
1. **Conversion**- is the formulation of a new proposition by interchanging the subject and predicate of an original proposition but leaving its quality unchanged.
 - **Kinds of Conversion**
 - a. Simple
 - b. Partial
 - **Note:**
 - **Convertend** - the original proposition.
 - **Converse** - the new proposition.
 - **Conversion** - the process itself.
 - a. **Simple Conversion**- the conversion is simple if the quantity of the converse is the same as the quantity of the convertend.
 - E to E
 - I to I
 - b. **Partial Conversion**- the conversion is partial if the quantity of the proposition is reduced from universal to particular.

-A to I
-E to O

○ **Rules for Conversion**

1. Interchange the subject and the predicate;
2. Retain the quality;
3. Do not extend any term.

○ **Note:**

1. Often it is advisable to reduce proposition to logical form.
2. Beware of converting A propositions by simple conversion.
3. O propositions cannot be converted.
4. The actual existence of a subject may not be asserted in the converse if it has not been asserted in the convertend.

○ **Examples of Simple Conversion:**

(E to E) No cat is a dog
No dog is a cat.
(I to I) Some houses are white.
Some white (things) are houses.

○ **Example of Partial Conversion:**

(A to I) All men are mortal.
Some mortal (beings) are men.
(E to O) No aliens are voters.
Some voters are not aliens.

2. Obversion- the formulation of a new proposition by retaining the subject and quantity of an original proposition, changing its quality, and using as predicate the contradictory of the original predicate.

- It also involves either the use or removal of two negatives: the use or omission of one negative changes the quality, the use or omission of the other negative changes the predicate to its contradictory.

○ **Note:**

- **Obvertend** - the original proposition.
- **Obverse** - the new proposition.
- **Obversion** - the process itself.

○ **Rules for Obversion**

1. Retain the subject and the quantity of the obvertend;
2. Change the quality. If the obvertend is affirmative, the obverse must be negative; and if the obvertend is negative to obverse must be affirmative;
3. As predicate, use the contradictory predicate of the original proposition.

○ **Examples of Obversion:**

(A to E) "Every cat is an animal" to
"No cat is a non-animal"
(E to A) "No cat is a dog" to
"Every cat is a non-dog"

(I to O)

"Some house is white" to

"Some house is not non-white"

(O to I)

"Some house is not white" to

"Some house is non-white"

3. Contraposition- is the formulation of a new proposition whose subject is the contradictory of the original predicate.

- **Kinds of Contraposition**

- a. Partial
- b. Complete

- **Note:**

- **Contraponend** - the original proposition.
- **Contraposit**- the new proposition.
- **Contraposition**- the process itself.

- **a. Partial Contraposition**

- **Rules for Partial Contraposition**

1. The subject is contradictory of the original predicate.
2. The quality is changed.
3. The predicate is the original subject.

- **To get the Partial Contraposition:**

1. Obvert
2. Converse the obverse

- **Symbols and their Partial Contraposition**

A to E	E to I
O to I	I no partial contraposit

- **Example of Partial Contraposition:**

A to E

(Contraponend) Every dog is an animal,

Obvert: No dog is a non-animal

Convert: No non-animal is a dog (Contraposit)

- **b. Complete Contraposition**

- **Rules for Complete Contraposition**

1. The subject is contradictory of the original predicate.
2. The quality is not changed.
3. The predicate is the contradictory of the original subject.

- **To get Complete Contraposition:**

1. Obvert
2. Converse the obverse
3. Obvert the converse of the obverse

- **Symbols and their Complete Contraposition**

A to A	E to O
O to O	I no complete contraposit

- **Example of Complete Contraposition:**

A to A

Contraponend) Every man is mortal,

Obvert: No man is non-mortal,

Convert: No non-mortal is man,

Obvert: Every non-mortal is a non-man (Contraposit)

4. Inversion - is the formulation of a new proposition whose subject is the contradictory of the original subject.

- **Kinds of Inversion**

- a. Partial
- b. Complete

- **Note:**

Invertend- the original proposition

Inverse- the new proposition

Inversion - the process itself

- **a. Partial Inversion**

- **Rules for Partial Inversion**

1. The quality is changed.
2. The predicate is the same as the original proposition.

- **To get Partial Inversion:**

1. Obvert
2. Convert
3. Obvert
4. Convert
5. Obvert

- **Symbols and their Partial Inversion** (Only A & E can be Inverted)

A to O

E to I

- **Example of Partial Inversion:**

A to O

(Invertend)

Obvert:

Convert:

Obvert:

Convert:

Obvert:

Some non-cat is not an animal. (Inverse)

Every cat is an animal,

No cat is a non-animal,

No non-animal is a cat,

Every non-animal is a non-cat,

Some non-cat is a non-animal,

- **b. Complete Inversion**

- **Rules for Complete Inversion**

1. The quality is not changed.
2. The predicate is the contradictory of the original predicate.

- **To get Complete Inversion:**

1. Convert
2. Obvert
3. Convert
4. Obvert

- **Symbols and their Complete Inversion** (Only A & E can be Inverted)

A to I

E to O

- **Example of Complete Inversion:**

E to O

(Invertend)	No cat is a dog,
Convert:	No dog is a cat,
Obvert:	Every dog is a non-cat,
Convert:	Some non- cat is a dog,
Obvert:	Some non-cat is not a non-dog. (Inverse)

HYPOTHETICAL PROPOSITION

Hypothetical Proposition

- A compound proposition that does not affirm nor negate the components themselves but only the nature and validity of their connection.
- Does not declare an unconditional affirmation or denial, but expresses a relation of dependence such as an opposition or a likeness between two clauses.

3 kinds of HYPOTHETICAL PROPOSITION

1. Conditional
2. Disjunctive
3. Conjunctive

1. Conditional Proposition

- Expresses a real dependence between a condition (antecedent) which is usually introduced by the word "IF", and the conditioned (consequent) introduced by the word "THEN".

Examples:

- If a man is farsighted, then he needs eyeglasses.
- If the rain stops, then we can go hiking.
- If the person is mute, then he can't communicate well.
- If jeepney drivers go on strike, then transportation service will be paralyzed.
 - From the given examples, we can observe that there is a real dependence between the antecedent and the consequent.
 - The truth of the consequent necessarily follows from the truth of the antecedent.
 - The truth of a conditional proposition depends on the conditional connection between the two components, prescinding from the truth value of the components taken separately.
- **The following examples, therefore, are not true:**
 - If Francis is the brother of James, then Francis is the child of James' father.
 - If you get operated on, then you'll survive.
 - Conditional proposition can be expressed in varied forms.
 - The statement, *"If he pays his income tax honestly, then he will not be penalized,"* can equally be expressed without "then": *If he pays his income tax honestly, he will not be penalized.*
 - At times, the order of the antecedent and the consequent can be reversed provided that the "if" still precedes the antecedent as in this example: *We will understand each other if we have a dialogue.*
 - Other words may be used for "if" like *in case, provided that, in the event that, given that, on the condition that,* etc.

Examples:

- In case he pays his income tax honestly, he will not be penalized.
- That he pays his income tax honestly implies that he will not be penalized.
- His paying the income tax honestly entails his not being penalized.

2. Disjunctive Proposition

- It is one whose subject or predicate consists of parts which exclude each other.
- It is sometimes called an EITHER OR statement due to its construction.
- The parts of a disjunction are called **disjuncts (alternants)**.
- Expresses alternatives (disjuncts) all of which cannot be true or false together but only one can be true, and the other must be false (exclusive disjunction), or at least one of which is true with the possibility that the rest of the alternatives may also be true (inclusive disjunction).

a. Exclusive Disjunction

- Here, the disjuncts are mutually exclusive, i.e., one necessarily excludes the other.
- One of the disjuncts is necessarily true, while all the others are false.

Examples:

- The vehicle is either moving or not moving.
- The request is either answered or unanswered.
- The patient was either dead or alive.
 - The truth of an exclusive disjunction depends on the mutual exclusion of the disjunct.
 - The exclusive nature of the disjuncts is due to the opposition between contradictory terms.

These examples, therefore, are not true: (exclusive disjunction)

- The man is either young or old.
- The applicant may either be the first or the last in line.

They should be expressed this way:

- The man is either young or not young.
- The applicant is either the first or not the first.

b. Inclusive Disjunction

- Here, at least one of the disjuncts is true, although the rest can also be true at the same time.
- The disjuncts can all be true but all of them cannot be false altogether.

Examples:

- You either have a mango or an apple.
- One can avail himself of the scholarship grant if he is either a Dean's Lister or a financially poor person.
- You can eat spaghetti or cake or fried chicken.
- I can sing or dance or smile.

3. Conjunctive Proposition

- One which denies that two contrary predicates together can be true of the same subject at the same time.
- It denies simultaneous possibility of two alternatives.

Examples:

- You cannot stand and sit at the same time.

- A man cannot walk forward and backward at the same time.
 - The truth of a conjunctive proposition depends on the incompatibility of the alternatives.

These examples, therefore, are not true: (conjunctive proposition)

- We cannot have rain and sunshine at the same time
- No one can sing and dance at the same time.

INFERENCE & SYLLOGISM

Reasoning

- A mental act whereby starting with several judgments which we relate to one another, we arrive at a new judgment which necessarily follows from the preceding ones.

Example:

All vehicles are instruments of transportation;
But jeep is a vehicle;
∴ Jeep is an instrument of transportation.

- Every reasoning process, therefore involves a number of previously known truths.
- These truths are known as PREMISES.
- The new truth is known as CONCLUSION.

A CATEGORICAL syllogism is an argument consisting of exactly three categorical propositions (2 premises and 1 conclusion), having three CATEGORICAL TERMS (Major, Minor, Middle), which are used exactly twice in the syllogism.

Inference

- It is drawing of a conclusion from one or more premises.
- It is a mental process by which we reach a conclusion based on specific evidence.
- It is the stock and trade of criminal investigators examining crimes, of doctors diagnosing diseases, and of car mechanics repairing engine problems.

Kinds of Inference

- **IMMEDIATE** inference is when the conclusion is drawn from only one premise.
- **MEDIATE** inference is when the conclusion is drawn from two premises.

Syllogism

- The mental product of inferential thinking is ARGUMENT.
- The external expression of an argument is SYLLOGISM.

2 Kinds of Syllogism

1. CATEGORICAL syllogism is an inferential thinking that draws a conclusion in an absolute manner.

Example:

Law implies order;
But the ten commandments are laws;
∴ The ten commandments implies order.

2. **HYPOTHETICAL syllogism** is an inferential thinking which concludes with certainty, affirming or denying a statement from the affirmation or denial of another.

Example:

If the students cheat, then they should be punished;
But the students cheat;
∴ They should be punished.

Constituents of Categorical Syllogism

M - Middle Term

P - Major Term

S - Minor Term

p - Particular

u - Universal

+ - Affirmative proposition (positive sign)

- - Negative proposition (negative sign)

CATEGORICAL SYLLOGISM

Categorical Syllogism- is a deductive argument consisting of 3 categorical propositions which contain exactly 3 terms.

Example:

- All Filipinos are hospitable persons
- All Ilocanos are Filipinos
- Therefore All Ilocanos are hospitable persons

How to identify the terms in the syllogism?

1.) Major Term (P)

- Found in the major premise as either the subject or the predicate.
- The predicate of the conclusion.

*In the example above, the major term is “hospitable persons” because it is the predicate of the conclusion.

2.) Minor Term (S)

- Found in the minor premise as either the subject or predicate
- The subject of the conclusion.

*In the example above, the minor term is “Ilocanos” because it is the subject of the conclusion

3.) Middle Term (M)

- The only term which is not found in the conclusion and the only term which appears twice in the premises.

*In the example above, the middle term is “Filipinos” because it is found in the two premises.

Example:

- ✦ All lawyers are politicians.
- ✦ Some lawyers are liars
- ✦ Therefore some liars are not politicians

- Major term: politicians
- Minor term: liars
- Middle term: lawyers

General Rules for a Valid Syllogism

1. There should be exactly 3 terms to be used throughout the whole argument.

- **Violation: fallacy of 4 terms (equivocation)**

Example:

- ✖ A ruler measures 12 inches.
- ✖ But GMA is a ruler.
- ✖ Therefore, GMA measures 12 inches.

* This syllogism is invalid because a word is used in two different propositions with different meaning.

* This syllogism committed the fallacy of equivocation or fallacy of 4 terms.

2. The middle term must be distributed at least once in the premises.

- **Violation: Fallacy of the undistributed middle term.**

Examples:

- ✖ All men are unique beings.
- ✖ Some unique beings are red.
- ✖ Therefore, some red beings are men.

- ✖ All dogs are animals.
- ✖ All cats are animals.
- ✖ Therefore, all cats are dogs

* These syllogisms are invalid syllogisms because the middle terms “unique beings” and “animals” are undistributed in the two premises.

*These syllogisms committed the fallacy of Undistributed Middle Term.

3. If the term is distributed in the conclusion then the same term must also be distributed in the premises.

- **Violation: Fallacy of the illicit major term or fallacy of illicit minor term.**

Example:

- ✖ All cats are animals.
- ✖ No bats are cats.
- ✖ Therefore, No bats are animals.

- ✖ All CICians are students.
- ✖ No Ateneans are students.
- ✖ Therefore, no Ateneans are students.

- ✖ All criminals are law-violators.
- ✖ All criminals are evil persons.
- ✖ Therefore, all evil persons are law-violators.

*These syllogisms are invalid syllogisms. The first and second example committed the fallacy of illicit major term and the third example committed the fallacy of illicit minor term.

1. **Illicit Major** means the predicate is universal in the conclusion not in the major premise.

2. **Illicit Minor** means that the subject is universal in the conclusion but not in the minor premise.

4. **Two negative premises are not allowed.**

- **Violation: Fallacy of exclusive terms**

Example:

- ✖ Rizza is not a joyful person
- ✖ Mae is not Michelle.
- ✖ Therefore, Mae is not a joyful person.

*This syllogism is invalid syllogism and had committed the fallacy of exclusive terms.

5. **One of the negative premises is allowed if and only if the conclusion is negative.**

- **Violation: Fallacy of drawing an affirmative conclusion based from a negative premise.**

Example:

- ✖ All teachers are models.
- ✖ Some teachers are not wise persons.
- ✖ Therefore, some wise persons are models.

* This syllogism is invalid and committed the fallacy of drawing out an affirmative conclusion based from a negative premise.

6. **No particular conclusion can be derived from 2 universal premises.**

- **Violation: Existential fallacy**

7. **Two affirmative premises cannot give a negative conclusion.**

8. **When one premise is particular, the conclusion must be particular.**

9. **When both premises are particular, there is no conclusion.**

FIGURES & MOODS

The **figure** of a syllogism is determined by the location of the middle term, the term that appears in both premises.

The **moods** refer to the different possible combinations of premises and conclusion according to the quantity and quality.

The **figure** of a syllogism is something to do with the position of the middle term in the premises.

Four figures of Categorical Syllogism

- FIGURE 1 - SUB-PRE
- FIGURE 2 – PRE-PRE
- FIGURE 3 – SUB-SUB
- FIGURE 4 – PRE-SUB

FIGURE 1 - SUB-PRE- the middle term is the subject of the major premise and the predicate of the minor premise.

Example:

Prisoners are law- offenders;
But, some criminals are prisoners;

∴ Some criminals are law-offenders.

FIGURE 2 - PRE-PRE -the middle term is the predicate of both premises.

Example:

Quails are not migratory birds;

Robins are migratory birds;

∴ No robins are quails.

FIGURE 3- SUB-SUB- the middle term is the subject of both premises.

Example:

Impulses are unreasonable actions;

Some impulses are regrettable;

∴ Some unreasonable actions are regrettable.

FIGURE 4- PRE-SUB- the middle term is the predicate of the major premise and the subject of the minor premise.

Example:

Fandango is a lively Spanish dance;

A lively Spanish dance is accompanied by a guitar and castanets;

∴ Fandango is accompanied by a guitar and castanets.

*The first figure has been considered to be the perfect syllogism because it is the way we tend to make arguments normally and naturally.

*The three other figures however are correct forms of syllogistic reasoning, even if they seem to be somewhat artificial and unnatural.

Moods-refers to the different arrangement of the four types of propositions (AEIO) in the syllogism.

- ✦ There will be 64 possible combinations.
- ✦ Now, 4 figures having 64 possible combinations (4*64) will give us 256 moods.
- ✦ Among these, there are only 19 valid moods, distributed to 4 figures.

MOODS

FIGURE 1 – AAA; EAE; AII; EIO

FIGURE 2 – EAE; AEE; EIO; AOO

FIGURE 3 – AAI; IAI; AII; EAO; OAO; EIO

FIGURE 4 – AAI; AEE; IAI; EAO; EIO

VARIETIES OF CATEGORICAL SYLLOGISM

KINDS

1. Enthymeme
2. Epichereme
3. Polysyllogism
4. Sorites
5. Dilemma

ENTHYMEME

-comes from the Greek terms “en thymos” which means “in the mind.”

-is an abridged syllogism in which one of the premises or conclusion is not expressed but implied or understood in the mind.

There are 3 Kinds:

1. Of the First Order
2. Of the Second Order
3. Of the Third Order

1. Enthymeme of the 1st Order

-major premise is omitted.

Example:

God is the supreme good; Hence, God is infinitely lovable.

The supreme good is infinitely lovable.

But, God is the supreme good.

Hence, God is infinitely lovable.

2. Enthymeme of the 2nd Order

-minor premise is omitted.

Example :

Cancers are life-threatening. Hence, Leukemia is life-threatening.

Cancers are life-threatening.

But, leukemia is cancer.

Hence, leukemia is life-threatening.

3. Enthymeme of the 3rd Order

-conclusion is omitted.

Example:

Anything worth pursuing is worth our time. But truth is worth pursuing.

Anything worth pursuing is worth our time.

But, truth is worth pursuing.

Hence, truth is worth our time.

Epichereme

-comes from the Greek word "epi cheir" which means "on hand."

-syllogism in which one or both premises are supported by proofs.

-accompanying proof is the explanation of the given premise.

2 Kinds:

1. Simple
2. Compound

1. Simple

-only one premise is accompanied by a proof or explanation.

Example:

Love is unexplainable for it is mysterious.

But, God is love.

Hence, God is unexplainable.

Anything mysterious is unexplainable.

But, love is mysterious.

Hence, love is unexplainable.

Compound

-both premises are accompanied by a proof or explanation.

Example:

Whatever spiritual is immortal because a spiritual being is incorruptible.

But, the human soul is spiritual because it is capable of grasping immaterial forms.

Hence, the human soul is immortal.

Whatever is spiritual is immortal because a spiritual being is incorruptible.

Anything incorruptible is immortal.

But, a spiritual being is incorruptible.

Hence, whatever is spiritual is immortal.

The human soul is spiritual because it is capable of grasping immaterial forms.

Anything capable of grasping immaterial forms is spiritual.

But, the human soul is capable of grasping immaterial forms.

Hence, the human soul is spiritual.

Polysyllogism

- a syllogism which is composed of several syllogisms so related to one another that the conclusion of one serves as the major premise of the other.

Example:

The more one is close to God, the more one suffers.

The more one suffers, the more one understands life.

The more one understands life, the more one relates to people.

The more one relates to people, the more one understands himself/herself.

Ergo, the more one is close to God, the more one understands himself/herself.

Sorites

-abridged form of polysyllogism wherein the intermediate conclusions are left out.

2 KINDS

1. Aristotelian Sorites

2. Goclenian Sorites

Aristotelian Sorites

-the predicate of the preceding premise becomes the subject of the following.

Example:

All A is B
All B is C
All C is D
All D is E
Ergo, All A is E.

Goclenian Sorites

-the subject of the preceding premise becomes the predicate of the following. In the conclusion, the subject of the last premise is united with the predicate of the first.

Example :

All A is B
All C is A
All D is C
All E is D
Ergo, All E is B.

Dilemma

-comes from the Greek dis (twice) and lemma (assumption).
-also known as horned argument.
-it is a form of argument whose major premise is a disjunctive or conditional proposition, and whose minor premise shows that whichever alternative the opponent chooses, is conclusively against him/her.

Kinds of Dilemma

1. Simple Constructive
2. Complex Constructive
3. Simple Destructive
4. Complex Destructive

1. Simple Constructive

Example :

Either you were asleep or you were not;
If you were, you deserve punishment (for dereliction of duty).
If you were not, you deserve punishment (for being incompetent).
Ergo, in either case, you deserve punishment.

2. Complex Constructive

Example:

If a candidate is a re-electionist, then she has nothing new to offer.

If a candidate is not a re-electionist, then she is not yet equipped in political matters.

But, she is either a re-electionist or not.

Therefore, in either case, she has nothing new to offer or she is not yet equipped in political matters.

3. Simple Destructive**Example:**

If Mario studies Medicine, he will have to study really hard and he will have to pay high tuition fees.

But, Mario either will not study really hard or he will not be able to pay high tuition fees.

Therefore, Mario will not study Medicine.

4. Complex Destructive**Example:**

If medicine prices are to be stabilized, then the government must have control over the stocks of pharmaceutical drugs.

If greater pharmaceutical drugs production is to be realized, the government should encourage greater cooperation among pharmaceutical industries.

But the government either does not have control over the stocks of pharmaceutical drugs or does not encourage greater cooperation among pharmaceutical industries.

Ergo, medicine prices are not stabilized or greater pharmaceutical drugs production is not realized.

HYPOTHETICAL SYLLOGISM**Definition**

-is one whose major premise is a hypothetical proposition while its minor premise and conclusion are categorical propositions.

-there is no major, minor, or middle term proper in the hypothetical syllogism.

3 Kinds of Syllogism**1. Conditional Syllogism**

-is one whose major premise is a conditional proposition consisting of an antecedent and consequent, while the minor premise affirms or denies either the antecedent or the consequent of the major premise, and the conclusion merely expresses whatever follows from the affirmation or denial.

-both the minor premise and the conclusion are categorical proposition.

Rules

1. Posit the antecedent, posit the consequent.

2. Negate the consequent, negate the antecedent.

3. Posit the consequent, no conclusion.
4. Negate the antecedent, no conclusion.

1. Posit the antecedent, posit the consequent.

- This is known, in symbolic logic as MODUS PONENS. If the antecedent is affirmed in the minor premise, the consequent must also be affirmed in the conclusion.
- The truth of the consequent follows from the truth of the antecedent.

Example:

If someone wins in the million-peso lotto draw, he becomes a millionaire;

But, Nikki wins in the million-peso lotto draw:

∴ Nikki becomes a millionaire.

2. Negate the consequent, negate the antecedent.

- This is known, in symbolic logic as MODUS TOLLENS.
- If the consequent is rejected in the minor premise, the antecedent must also be rejected in the conclusion.
- The falsity of the antecedent follows from the falsity of the consequent.

Example:

If someone wins in the million-peso lotto draw, he becomes a millionaire;

But, Nikki does not become a millionaire

∴ Nikki does not win in the million-peso lotto draw.

3. Posit the consequent, no conclusion.

- This rule gives rise to the fallacy of affirming the consequent that is committed when the consequent is affirmed in the minor premise.

Example:

If someone wins in the million-peso lotto draw, he becomes a millionaire;

But, Nikki becomes a millionaire;

∴ Nikki wins in the million-peso lotto draw.

4. Negate the antecedent, no conclusion.

- This rule gives rise to the fallacy of rejecting the antecedent that is committed when the antecedent is rejected in the minor premise.

Example:

If someone wins in the million-peso lotto draw, he becomes a millionaire;

But, Nikki didn't win the million-peso lotto draw;

∴ Nikki didn't become a millionaire.

Exception

-there are few cases in which one can conclude certainty (a) from the truth of the consequent to the truth of the antecedent, and (b) from the falsity of the antecedent to the falsity of the consequent.

-but this is due not to the legitimate nature of the conditional inferential procedure employed but to the formal necessity involved in the matter in the conditional major premise.

Example:

If this month is December, next month is January;
But, this month is not December;
∴ Next month is not January.

-this syllogism is a valid syllogism.

-the major premise involves sequential correlatives.

-in this case, even if the antecedent is rejected in the minor premise, the conclusion is still true.

Example:

If you failed in the test, you did not pass;
But, you did not pass;
∴ You failed.

-this syllogism is also a valid syllogism.

-the major premise expresses mutually exclusive alternatives.

-in this case even if the consequent is affirmed in the premise, the conclusion still true.

2. Disjunctive Syllogism

-is one whose major premise is a disjunctive proposition consisting of alternatives (disjuncts), while the minor premise affirms or denies any of disjuncts, and the conclusion merely expresses whatever follows from its affirmation or denial.

2 kinds of disjunctive syllogism:

Complete

Incomplete

- **Complete disjunction** - whose parts are mutually exclusive.

Rules: Complete Disjunction

1. Posit one alternative, negate the other. (Ponendo-tollens)
2. Negate one alternative, posit the other. (Tollendo-ponens)

1. Posit one alternative. Negate the other. (Ponendo-tollens)

- The minor premise affirms one disjunct and the conclusion rejects the other.

Examples

The accused is either guilty or not guilty;
But, he is not guilty; (affirmed)
∴ He is not guilty. (rejected)

The accused is either guilty or not guilty;
But, he is guilty; (affirmed)
∴ He is guilty; (rejected)

2. Negate one alternative, posit the other. (Tollendo-ponens)

- The minor premise rejects one disjunct and the conclusion affirms the other.

Examples

The military operation in Basilan is either successful or unsuccessful;
But it is not successful; (rejected)
∴ It is unsuccessful. (affirmed)

The military operation in Basilan is either successful or unsuccessful;
But it is not unsuccessful; (rejected)
∴ It is successful. (affirmed)

- **Incomplete disjunction** - whose parts are not mutually exclusive.

Rules: Incomplete disjunction

1. Posit one alternative, sublate the other. (Ponendo-tollens)
2. Sublate one alternative, no conclusion.

1. Posit one alternative, negate the other. (Ponendo-tollens)

- The minor premise affirms one disjunct and the conclusion rejects the other.

Examples:

The test is either easy or difficult;
But it is easy; (affirmed)
∴ It is not difficult. (rejected)

The test is either easy or difficult;
But it is difficult; (affirmed)
∴ It is not easy. (rejected)

Examples above follow the rule of contrariety. Easy and difficult are contraries. If one is true then one must be false. But one is false the other is doubtful.

2. Negate one alternative, no conclusion.

- The minor premise rejects one disjuncts and the conclusion affirms the other.

Examples:

The test is either easy or difficult;
But it is not easy; (rejected)
∴ It is difficult. (affirmed)

This machine is either light or heavy;
But it is not heavy; (rejected)
∴ It is light. (affirmed)

That syllogism (example #2) is invalid. The machine maybe moderate in weight.

3. Conjunctive Syllogism

-is one whose major premise is a conjunctive proposition consisting of alternatives (conjuncts), while the minor premise affirms or denies any conjuncts, and the conclusion merely expresses whatever follows from its affirmation or denial.

RULES

1. Posit one alternative, negate the other. (Ponendo-tollens)
2. Negate the alternative, no conclusion.

1. Posit one alternative, negate the other. (Ponendo-tollens)

- The premise affirms one conjunct and the conclusion rejects the other.

Example:

The passenger cannot be in the tricycle and in the bus at the same time;

But, he is in the tricycle ;(affirmed)
∴ He is not on the bus. (rejected)

2. Negate one alternative, no conclusion.

-But from the falsity of one conjunct, the truth of the other does not necessarily follow.

Example:

The passenger cannot be in the tricycle and in the bus at the same time;

But, he is not in the tricycle;(rejected)
∴ He is in the bus.(affirmed)

-The conclusion, "He is in the bus" does not follow from the premise. There are vehicles other than bus that he can take. The fallacy is called the Fallacy of Tollendo- Ponens

FALLACY

Definition

- Comes from the Latin word **fallo** or **fallere** which means to deceive.
- It refers to any error in reasoning.
- May be formal or informal.

Formal- refers to errors in relation to the form or structure of reasoning or argument.

- a. The point in question is not whether a conclusion is true or false but whether the form of argument is correct or incorrect, valid or invalid.

Informal- refers to errors with regard to the content or the meaning of the argument.

- b. It is committed when factors contributing to reasoning are distorted.

Some Informal Fallacies which are frequently committed

1. Argumentum Ad Misericordiam (Appeal to Pity or Sympathy)

- c. This is the error of by-passing logic and the point at issue by appealing to pity or sympathy.

Example:

Sir, maawa napo kayo sa akin, Ipasa nyo na po ako! May tatlong anak po akong pinapag-aral at sinusuportahan."

2. Argumentum Ad Populum (Appeal to the People)

This is committed when one evades the point under discussion and instead appeals to the opinions, passions or prejudices of the people.

Example:

Migz Zubiri is sure to win the senatorial position because he has been consistently topping election surveys.

3. Argumentum Ad Hominem (Argument Against the Person)

This happens when the person of the arguer is attacked, rather than his argument.

Examples:

“Wag ninyong pinaniniwalaan yung sinasabi ni Peter, pasaway, bolero at tsaka hindi maganda ang background ng pamilya niya.”

“Mali ang sinabi nya dahil estudyante lang siya.”

4. Argumentum Ad Baculum (Appeal to Force or Threat)

This is committed when one seeks to establish a conclusion by resorting to force or intimidation.

Examples:

“Kapag hindi ka nakuha sa santong dasalan, kukunin kita sa santong paspasan.”

“Kapag hindi mo ako minahal, hindi kita ipapasa.”

5. Argumentum Ad Vericundiam (Appeal to wrong authority)

This happens when a wrong or unqualified authority is cited to establish a conclusion.

Example:

Another killer earthquake will hit Cabanatuan City within three years, Sarah Geronimo said.

6. Argumentum Ad Ignorantiam (Appeal to Ignorance)

This occurs when one proves that a thing is true because it cannot be proven false or a thing is false because it cannot be proven true.

Example:

“Wala talagang Diyos kase hindi ko siya nakikita kahit na sa microscope.”

7. Addendum: Non sequitur (It does not follow)

An argument whose conclusion does not follow logically from the premises as when two ideas are disconnected.

Example:

The sky is blue, therefore I love you.

8. Post hoc ergo propter hoc (False Cause)

Example:

The Basketball Varsity of the College of the Immaculate Conception won the championship game because the players drank Milo Energy drink before the game.

9. Complex Question

This is asking a question that presupposes an answer to another question that has not been asked and answered.

Example:

a. “Is he your 3rd boyfriend?”

**ARGUMENT
Definition**

- Is an attempt to persuade someone of something, by giving reasons or evidence for accepting a particular conclusion.

- The general structure of an argument in a natural language is that of premises (typically in the form of proposition, statements or sentences) in support of a claim: the conclusion.

Informal arguments

- As studied in *informal logic*, are presented in ordinary language and are intended for everyday discourse.

Example

Geological events produce rock.
Rock is a type of music.
Geological events produce music

Formal Argument

- As studied in *formal logic* (historically called *symbolic logic*, more commonly referred to as *mathematical logic* today) and are expressed in a formal language.

Example

- All women are brilliant.
- I am a woman.
- Therefore, I am brilliant.

Deductive Arguments

- Is an argument in which the arguer claims that it is impossible for the conclusion to be false given that the premises are true. (Involves necessary reasoning)
- The premises are intended to provide support for the conclusion that is so strong that, if the premises are true, it would be *impossible* for the conclusion to be false.

Truth preserving arguments: Deductive Arguments

Examples

- There are 32 books on the top-shelf of the bookcase, and 12 on the lower shelf of the bookcase. There are no books anywhere else in my bookcase. Therefore, there are 44 books in the bookcase.
- Bergen is either in Norway or Sweden. If Bergen is in Norway, then Bergen is in Scandinavia. If Bergen is in Sweden, then Bergen is in Scandinavia. Therefore, Bergen is in Scandinavia.

Kinds of Deductive Arguments

- Valid Deductive Argument
- Invalid Deductive Argument
- Sound Argument

Valid Deductive Argument

- Impossible for the conclusion to be false given that the premises are true.

Example

All monkeys are primates
All primates are mammals
So, all monkeys are mammals

Invalid Deductive Argument

- It is possible for the conclusion to be false given the premises are true.

Example

- If Sue misses her plane she will be late for the conference.
- Sue is late for the conference.
- Therefore, she missed her plane.

Sound Argument

- A valid argument and has all true premises.

Example

- All residents of Fenton are untrustworthy
- Wayne lives in Fenton
- Wayne is untrustworthy

Inductive Argument

- Is an argument in which the arguer claims that it is probable that the conclusion be false given that the premises are true. (Probabilistic reasoning)
- The premises are intended only to be so strong that, if they are true, then it is *unlikely* that the conclusion is false.

Example:

- It has snowed in Massachusetts every December in recorded history.
- Therefore, it will snow in Massachusetts this coming December.
- The meerkat is closely related to suricat.
- The suricat thrives on beetle larvae.
- Therefore, probably the meerkat thrives on beetle larvae.

Kinds of Inductive Argument

- **Strong Inductive**
- **Weak Inductive**
- **Cogent Argument**

Strong Inductive

- Which it is improbable that the conclusion be false given that the premises are true.

Example:

This cooler contains 30 cans.

25 cans selected at random were found to be Pabst Blue Ribbon (PBR).

Probably all the cans are PBR.

Weak Inductive

- Conclusion does not follow probably from the premises, even though it is claimed to.

Example:

This cooler contains 30 cans.

3 cans selected at random were found to be PBR.

Probably all the cans are PBR.

Cogent Argument

- Inductive argument that is strong and has all true premises; if either condition is missing, the argument is uncogent.

Example

Swimming in the Caribbean is usually lots of fun. Today the water is warm, the surf is gentle, and on this beach there are no dangerous currents. Therefore, it would be fun to go swimming here now.

Other types of Argument

Argument from Analogy

- May be thought of as argument from the particular to particular.
- An argument that depends on the existence of an analogy, or similarity between two things or states of affairs

Examples:

- ✓ Plato was mortal, and Socrates was like Plato in other respects, then Socrates was mortal
- ✓ Someone might argue that because Christina's Porsche is a great handling car, it follows that Angela's Porsche must also be a great handling car.

Argument based on Mathematics

- Arguments in which the conclusion depends on some purely arithmetic or geometric computation or measurement.

Example:

Property P is true of the number 0.
For all natural numbers n , if P holds of n then P also holds of $n + 1$.
Therefore, P is true of *all* natural numbers.

Argument from Definitions

- In which the conclusion is claimed to depend merely upon the definition of some word or phrase used in the premise or conclusion.

Example:

Some might argue that because Claudia is mendacious, it follows that she tells lies, or that because a certain paragraph is prolix, it follows that it is exclusively wordy.

Argument from Authority

- Concludes something is true because an expert or witness has said that it is.

Example:

A person might argue that earning for Hewlett – Packard Corp. will be up in the coming quarter of a statement to that effect by an investment counselor.

Argument from Signs

- That proceeds from the knowledge of thing or situation that the sign symbolizes.
- "Sign" means any kind of message (usually visual) produced by an intelligent being.

Example:

When driving on an unfamiliar highway one might see a sign indicating that the road makes several sharp turns one mile ahead.

Argument Against the Person

- Always involves two arguers.

- One of them advances (either directly or indirectly) a certain argument and the other then responds by directing his/her attention not to the first person's argument but to the first person himself.

Example:

Bill: "I believe that abortion is morally wrong."

Dave: "Of course you would say that, you're a priest."

Bill: "What about the arguments I gave to support my position?"

Dave: "Those don't count. Like I said, you're a priest, so you have to say that abortion is wrong. Further, you are just a lackey to the Pope, so I can't believe what you say."

ETHICS

Definition

Ethics or moral philosophy is a branch of philosophy that involves systematizing, defending, and recommending concepts of right and wrong conduct.

The field of ethics, along with aesthetics, concerns matters of value, and thus comprises the branch of philosophy called axiology

Ethics seeks to resolve questions of human morality by defining concepts such as good and evil, right and wrong, virtue and vice, justice and crime. As a field of intellectual inquiry, moral philosophy also is related to the fields of moral psychology, descriptive ethics, and value theory.

Three Major Areas of Study of Ethics

1. **Meta-ethics**, concerning the theoretical meaning and reference of moral propositions, and how their truth values (if any) can be determined
2. **Normative ethics**, concerning the practical means of determining a moral course of action
3. **Applied ethics**, concerning what a person is obligated (or permitted) to do in a specific situation or a particular domain of action.

Theories of Ethics

1. Hedonism

- Gk. (hedone "pleasure" and hedys "sweet") is an ethical doctrine, which claims that pleasure is the norm of morality.
- Ang kasiyahan ay siyang sukatan ng moralidad.
- Pleasure in this context, is the satisfaction of desire; hence, the greater the pleasure, the better.

2. Stoicism

- Stoicism was one of the new philosophical movements of the Hellenistic period. The name derives from the porch (stoa poikilê) in the Agora at Athens decorated with mural paintings, where the members of the school congregated, and their lectures were held.
- As an ethical doctrine, STOICISM considers apathy or indifference to pleasure as the moral norm.

3. Epicureanism

- Founder: Epicurus (341-270 B.C.)
- Athenian Philosopher.
- Speaking of the noblest aim of philosophy, Epicurus pointed to Ataraxia, or peaceful and tranquil life.
- Professes moderate pleasure as the moral norm.
- It is the intermediate pleasure doctrine between hedonism and stoicism.
- Hedonism teaches an all-out pleasure norm of morality.
- Stoicism an indifference pleasure or no pleasure at all.
- Moderate pleasure. "Katamtaman lamang"
- What Epicurus meant by pleasure is not fleeting but permanent, or that state of deep peace and perfect contentment in which we feel secure against the storms of life.

4. Situationism

- Joseph Fletcher is an American Protestant medical doctor, situationism is also known as contextualism or situation ethics.
- He mentions the three approaches to morality:
 - Legalism
 - Antinomianism
 - Situationism

5. Humanistic Ethics

- Known under various names such as endemonism, perfectionism, self-realizationism, or self-actualizationism, humanistic ethics claims that self-realization is the true standard of morality.
- Man is being with others in the world, hence he cannot live the good life apart from others not only individual but also social in its emphasis and outlook.
- No person can fully develop himself independently of his fellowmen

6. Thomistic Ethics

- St. Thomas Aquinas (1225-1274).
- 13th century profound Italian philosopher and theologian.
- The source of moral law is reason itself.
- Reason recognizes the basic principle "Do good, avoid evil."
- "Gawin ang mabuti, iwasan ang masama."
- Right reason or the voice of reason (tinig ng katuwiran) as the moral norm.
- "The good must be done, and evil, avoided"

7. Mightism

- It professes that whenever an act enhances one's strength or capacity to rule.

8. Rational Ethics

- Merely an ethical system based on logic, on reason.

9. Yin- yang Polarity

- Based on a single principle that the universe is run by the Great Ultimate, or Tao.
- The philosophy of the Tao is that everything is divided into two principles that are opposite one another in their existence and actions, yin and yang.
- Balanced.

10. Filipino Centripetal Ethics

- Law of Karma

- Whatever you have planted, that you are going reap.

*****the end*****

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